

Quiz 06

Instructions: Answer the questions with precise definitions. Your definitions should either be stated exactly as in the lecture or textbook, or be clearly equivalent to them. Having the “right idea” but not a precise formulation will not receive credit.

Problem 1. What does it mean for the random variables $X_1, \dots, X_n$ to form a **random sample** from a distribution with probability density function f ?

Problem 2. Let $X_1, \dots, X_n$ be a random sample. What is the formula for the **sample variance** S^2 ?